

Model Rocket Vertical Launch Simulation

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1 Introduction

This report aims to explain the general physics and mathematics behind the powered flight of a rocket, the parameters used to describe the state of the rocket during flight, and the methods used to simulate its flight in Python.

2 Jet Propulsion

Rockets use jet propulsion to generate thrust and fly, and understanding the general physical concept behind jet propulsion is crucial for understanding how rockets work. In physics, there is a quantity that measures the amount of motion that a given body possesses. This quantity is called momentum:

$$\vec{p} = m\vec{v}$$

Where m is the inertial mass and $\vec{v}$ is the velocity vector. From the Newton's second principle we know that the net force acting on a body is:

$$\vec{F} = m\vec{a}$$

The acceleration is the first time derivative of the velocity vector:

$$\vec{F} = m \frac{d\vec{v}}{dt} = \frac{d\vec{p}}{dt}$$

The net force can be written as the first time derivative of the momentum. When we have a system of several bodies that interact with each other according

to the Newton's third principle the net vector sum of all forces in the system will be 0 so:

$$\sum_i \frac{d\vec{p}_i}{dt} = \frac{d \sum_i \vec{p}_i}{dt} = 0$$

$$\sum_i \vec{p}_i = \vec{p} = \text{const}$$

the vector sum of the momentum of all bodies is called momentum of the system and in this case is conserved. The conservation happens only when:

$$\vec{F}_{ij} = -\vec{F}_{ji}$$

all forces are internal so the net sum of all forces is 0. This is called conservation of mechanical momentum law and it is only valid when all forces are internal. When external forces are present in the system:

$$\sum_i \frac{d\vec{p}_i}{dt} = \frac{d \sum_i \vec{p}_i}{dt} = \frac{d\vec{p}}{dt} = \sum \vec{F}_e$$

Where $\sum \vec{F}_e$ is the vector sum of all external forces acting on the mechanical system. From here the differential change in the system's momentum is:

$$d\vec{p} = \sum \vec{F}_e dt$$

In order to find the velocity $\vec{v}$ of the rocket as function of t we will find the differential change in the momentum of the mechanical system : rocket-expelled gases from the engine. The movement of the rocket is only in vertical direction so the velocity, the momentum and the forces acting on the rocket have only

vertical components the x and z vector components are 0. When the rocket engine is running the vehicle expels mass in the form of exhaust gases so in our model we will use that the mass is expelled with constant rate:

$$\frac{dm}{dt} = \mu = \text{const}$$

Where $\mu < 0$ because the mass is decreasing. From here we get that $dm = \mu \cdot dt < 0$. At a moment t before from the rocket is expelled infinitesimal mass $|dm| = -dm$ the momentum of the system is:

$$\vec{p}(t) = m(t)\vec{v}(t)$$

After infinitesimal mass $-dm$ is expelled:

$$\vec{p}(t + dt) = (m(t) + dm)(\vec{v}(t) + d\vec{v}) - dm(\vec{u} + \vec{v}(t) + d\vec{v})$$

Where $\vec{u}$ is the velocity of the expelled mass relative to the rocket and $\vec{u} + \vec{v}(t) + d\vec{v}$ is the velocity of the expelled mass relative to the ground (inertial frame of reference).

$$d\vec{p} = \vec{p}(t + dt) - \vec{p}(t) = md\vec{v} - dm\vec{u}$$

$$md\vec{v} - dm\vec{u} = \sum \vec{F}_e dt$$

The external forces acting on the system are the gravitational force and the force of air resistance, the buoyant force is neglected:

$$md\vec{v} - dm\vec{u} = (m\vec{g} + \vec{f})dt$$

$$|\vec{f}| = \frac{1}{2}\rho \cdot SCv^2$$

The force of air resistance is directly proportional to the square of the velocity magnitude. It is also a function of the air density, cross-sectional area and

coefficient of air resistance C which depends on the shape,angle of attack:

$$\alpha = 0$$

(vertical flight) and the mach number:

$$M = \frac{|\vec{v}|}{c}$$

where $c = 340m/s$ is the speed of sound in the air. Because of the different regimes in flight: subsonic: $M < 1$,transonic: $M \approx 1$,supersonic: $M > 1$ and the complex dependence between the mach number and C in this model we will assume for simplicity that C depends only on the shape of the nose cone of the rocket. The force of air resistance is always opposing the movement of the rocket so the force vector is in the opposite direction of the velocity vector $\vec{v}$:

$$\vec{f} = -\frac{1}{2}\rho.SC|\vec{v}|^2.\frac{\vec{v}}{|\vec{v}|} = -\frac{1}{2}\rho.SC|\vec{v}|\vec{v}$$

The air density is also changing with the altitude and in this model we use that the density is exponential function of the altitude $y(t)$:

$$\rho = \frac{\rho_0}{e^{\frac{y(t)}{H}}}$$

Where $\rho_0 = 1.225$ is the air density at sea level and $H = 10000$ is referent height.

$$m\frac{d\vec{v}}{dt} - \frac{dm}{dt}\vec{u} = m\vec{g} - \frac{1}{2}\rho.SC|\vec{v}|\vec{v}$$

$$m \frac{d}{dt} \begin{bmatrix} 0 \\ v_y \\ 0 \end{bmatrix} - \frac{dm}{dt} \begin{bmatrix} 0 \\ -|\vec{u}| \\ 0 \end{bmatrix} = m \begin{bmatrix} 0 \\ -g \\ 0 \end{bmatrix} - \frac{1}{2} \rho \cdot SC |\vec{v}| \begin{bmatrix} 0 \\ v_y \\ 0 \end{bmatrix}$$

When we write the equation only for y-axis and divide by m :

$$a_y + \frac{1}{m} \frac{dm}{dt} |\vec{u}| = -g - \frac{1}{2m} \rho \cdot SC |v_y| v_y$$

$$|\vec{u}| = Isp \cdot g$$

Where Isp is quantity that characterizes the efficiency of the rocket engine, in our model the $Isp = 250s$. From here for the acceleration we get:

$$a_y = -g - \frac{1}{2m} \rho \cdot SC |v_y| v_y - \frac{1}{m} \frac{dm}{dt} Isp \cdot g$$

3 Python code

Firstly we write down all parameters that we are going to use such as the dry and the fuel mass of the rocket, air resistance coefficients, the free-fall acceleration, the Isp of the rocket motor, etc. To solve the differential equations numerically for the acceleration and velocity we define function that contains our differential equations. To do this we define vector $\vec{S}$:

$$\vec{S} = (y, v_y, m)$$

The time derivative of this vector contains all the differential equations for the y-coordinate (altitude), for the velocity and for the mass. We write the equations

for the velocity, acceleration and rate of mass loss in the 'return' statement:

$$v_y = \frac{dy}{dt}$$

$$a_y = \frac{dv_y}{dt} = -g - \frac{1}{2m} \rho \cdot SC |v_y| v_y - \frac{1}{m} \frac{dm}{dt} I_{sp} \cdot g$$

$S = \pi \cdot r^2$ (cross-sectional area) and in the 'return' statement for the mass derivative while the engine is running we get:

$$\frac{dm}{dt} = \mu$$

and while the engine is not working:

$$\frac{dm}{dt} = 0$$

Then we set our initial conditions. For the time we create a NumPy array and for the numerical solution of all three equations we use `odeint` from `scipy.integrate`. The values for the mass derivative are controlled by 'if' statement that determines if the time for the completion of the burn has passed. Another 'if' statement controls the value of the air resistance coefficient because when the rocket begins to fall back down the geometry of the aft section is different from the geometry of the nose cone which leads to greater air resistance and when the parachute is open the resistance and the cross-section become greater. For the visualization of the functions we use `pyplot` from `matplotlib`. All function graphs are animated using `FuncAnimation` from `matplotlib.animation`. The number of the frames are calculated via the function `len()` that finds the length of the time array.

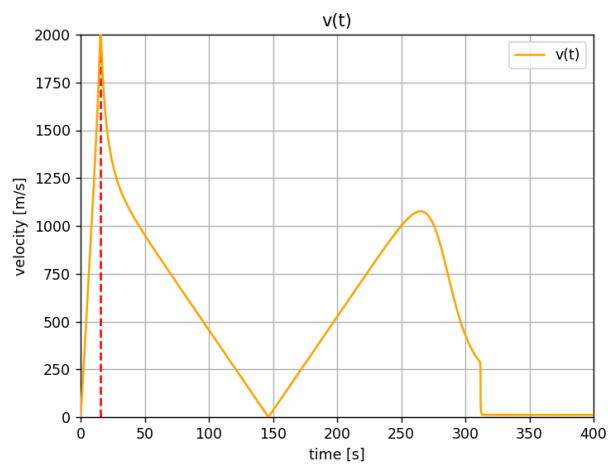
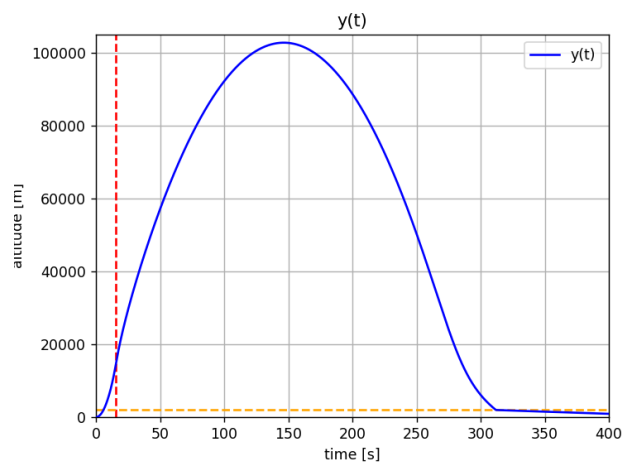


Figure 1: $y(t)$ and $v(t)$